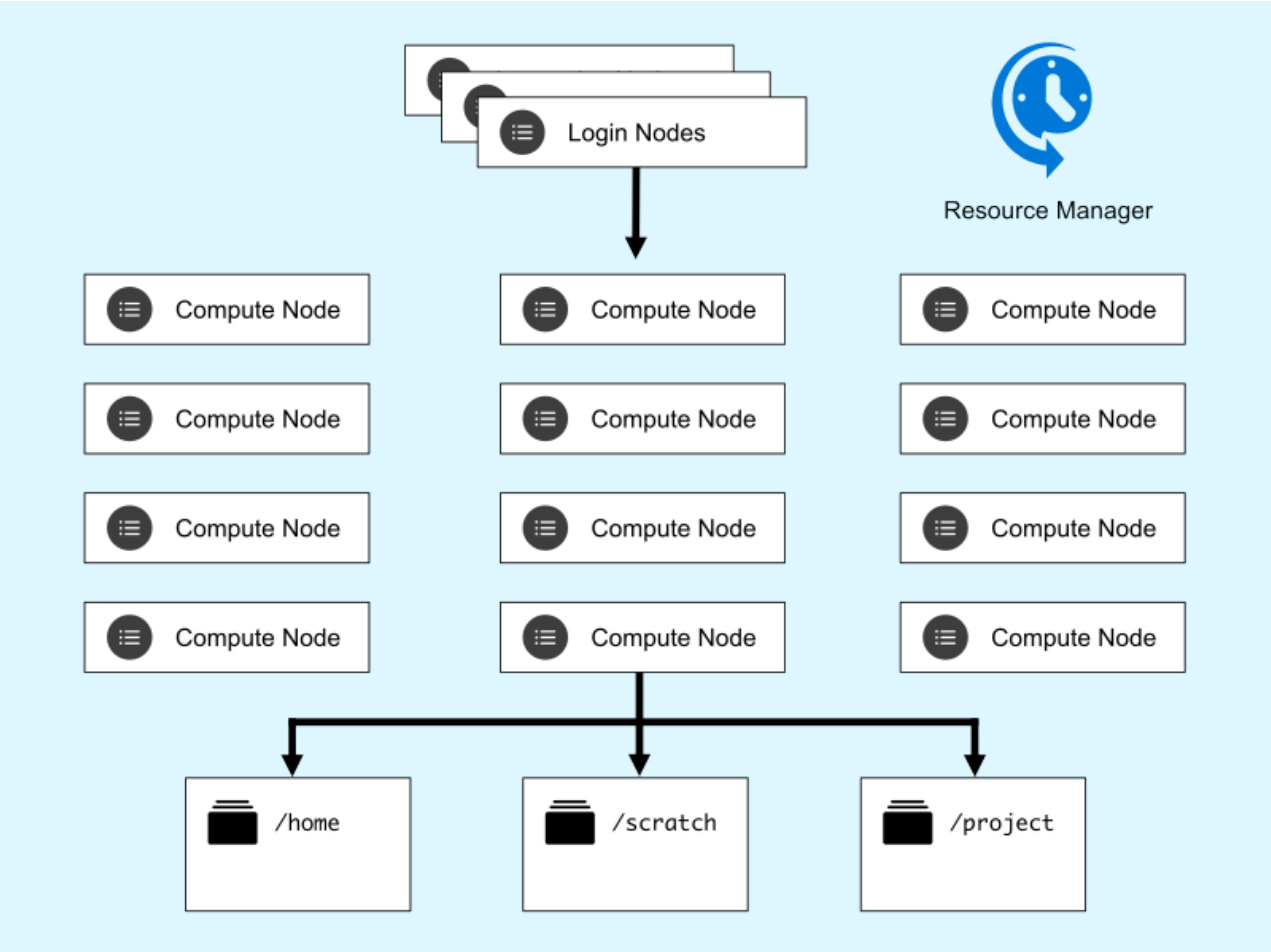
Accessing R on UVA HPC

Rivanna & Afton

- Rivanna and Afton are University of Virginia's High-Performance Computing (HPC) systems.



Rivanna/Afton

- Rivanna and Afton are the names for the University of Virginia's High-Performance Computing (HPC) systems.

- Uses environment modules (http://modules.sourceforge.net) to make software available to users.

- Overview : https://www.rc.virginia.edu/userinfo/hpc/.

- Besides lots of important information related to computing at UVA, you can sign up for a "Intro to HPC" session at the "Overview" link.

What is R? What is RStudio?

- Used to refer to both the programming language and the software that interprets the scripts written using it.
- RStudio is currently a popular way to write R scripts/commands but also to interact with the R software.
- R and RStudio on Rivanna: https://www.rc.virginia.edu/userinfo/hpc/software/r/

Why learn R?

- R does not involve lots of pointing and clicking, and that’s a good thing
- R code is great for reproducibility
- R is extensible
- R works on datasets of all shapes and sizes
- R can produce high-quality graphics
- R has a large community
- It is free, open-source and cross-platform

Some resources to learn R further

- R for beginners (https://cran.r-project.org/doc/contrib/Paradis-rdebuts_en.pdf)
- The R Manuals (https://cran.r-project.org/manuals.html)
- R for data science (https://r4ds.had.co.nz)
- Practical data science for stats (https://peerj.com/collections/50-practicaldatascistats)
- Efficient R programming (https://csgillespie.github.io/efficientR/index.html)
- https://data.library.virginia.edu/training/

https://uvacobi.github.io/genomics-r-intro/index.html

Beta

Learner View

Intro to R for Genomics Key Points Glossary Learner Profiles More

Search the All In One page

%

Collapse

EPISODES

Next: Introducing R...

Summary and Setup

Welcome to R! Working with a programming language (especially if it's your first time) often feels intimidating, but the rewards outweigh any frustrations. An important secret of coding is that even experienced programmers find it difficult and frustrating at times – so if even the best feel that way, why let intimidation stop you? Given time and practice* you will soon find it easier and easier to accomplish what you want.

Why learn to code? Bioinformatics – like biology – is messy. Different organisms, different systems, different conditions, all behave differently. Experiments at the bench require a variety of approaches – from tested protocols to trial-and-error. Bioinformatics is also an experimental science, otherwise we could use the same software and same parameters for every genome assembly. Learning to code opens up the full possibilities of computing, especially given that most bioinformatics tools exist only at the command line. Think of it this way: if you could only do molecular biology using a kit, you could probably accomplish a fair amount. However, if you don't understand the biochemistry of the kit, how would you